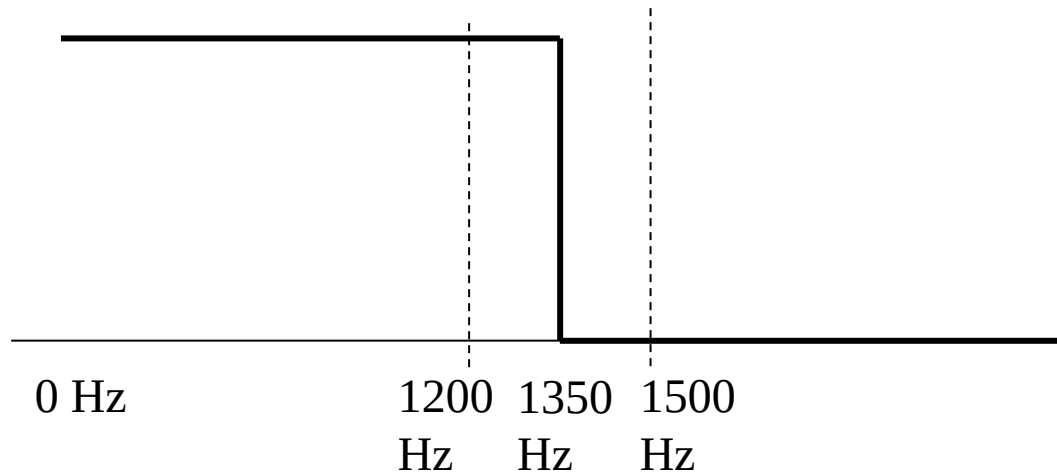


Homework 1 (Due: March 20th)

(1) Design a Mini-max **lowpass** FIR filter such that (40 scores)

- ① Filter length = 17, ② Sampling frequency $f_s = 6000\text{Hz}$,
- ③ Pass Band 0~1200Hz ④ Transition band: 1200~1500 Hz,
- ⑤ Weighting function: $W(F) = 1$ for passband, $W(F) = 0.6$ for stop band .
- ⑥ Set $\Delta = 0.0001$ in Step 5.



※ The code should be handed out by NTUCool, too.

Show (a) the frequency response, (b) the impulse response $h[n]$, and (c) the maximal error for each iteration.

- (2) How do we implement $y[n] = x[n] * (0.8^n u[n] + 0.5^n u[n])$ efficiently where $*$ means convolution and $u[n]$ is the unit step function? (10 scores)
- (3) (a) What are the two main advantages of the Fourier transform (FT)? (b) What are the two main problems to implement the FT? (10 scores)
- (4) Suppose that $x[n] = y(0.002n)$ and the length of $x[n]$ is 2000. If $X[m]$ is the FFT of $x[n]$, which frequencies do (a) $X[200]$ and (b) $X[1600]$ correspond to? (10 scores)
- (5) Why (a) the step invariance method and (b) the bilinear transform can reduce or avoid the aliasing effect in IIR filter design? (10 scores)
- (6) (a) Which of the following filters are usually even? (b) Which of the following filters are usually odd? (i) Notch filter; (ii) highpass filter; (iii) edge detector; (iv) integral; (v) differentiation 4 times; (vi) particle filter; (vii) matched filter. (10 scores)

(7) Use the MSE method to design the 7-point FIR filter that approximates the highpass filter of $H_d(F) = 1$ for $|F| < 0.25$ and $H_d(F) = 0$ for $0.25 < |F| < 0.5$.
(15 scores)

(Extra): Answer the questions according to your student ID number.
(ended with 0, 1, 2, 3, 5, 6, 7, 8)